

PIA 405 Course Outline

PIA 405 – Cost-Effectiveness and Non-Monetised Appraisal

Appraising what you cannot price, without pretending you can.

School	04 – Public Investment
Hours	20 (indicative band 18-22)
Version	1.0 (2026-09-05)
Model	AICTEM PIAS, D:/EDIP/models/PIASModelling
Reference country	Utopia (UTP), UTD 2024 prices
Prerequisites	PIA 403 for discounting and present values, and PIA 404 for the economic cost side. This course uses the same converted cost profile and stops before the benefit valuation that PIA 404 performs.
Audience	Health, education, justice and security sector analysts; appraisal economists working on projects whose outcome has no defensible money value; and reviewers who have to decide whether a benefit-cost ratio was the right instrument
Pass mark	60 %

What this course is

Some outcomes cannot be priced. Not because nobody has tried, but because the attempt would require a money value for something on which there is no defensible market, no willingness-to-pay study that transfers, and no policy consensus. Access to justice is one. So, in many contexts, is a year of primary schooling or an averted case of a disease with no burden estimate.

The wrong response is to look for a benefit to monetise until the ratio clears the hurdle. That is the reverse of appraisal. The right response is to change the instrument: compare the cost of achieving a defined outcome across alternatives, without requiring the outcome to be converted to money.

This course teaches cost-effectiveness analysis as PIAS implements it: the cost-effectiveness ratio, the incremental ratio and its four quadrants, least-cost comparison, and the discipline that both the discounted and undiscounted denominators are computed and stored because the choice changes the ratio by a factor of two over a thirty-year appraisal.

The register supplies its own worked case. Exactly one of the eleven projects is flagged `is_non_monetised = 1`: Regional Justice Centres, whose benefit-cost ratio is 0.00 and whose

first-year rate of return is NOT AVAILABLE, because it has no monetised benefit in any year. It is not a failed appraisal. It is a project appraised on the wrong instrument.

What a graduate can do

Decide whether an outcome can be defensibly monetised, and say what would have to be true for it to be; compute a cost-effectiveness ratio on both discounted and undiscounted denominators and explain why both are stored; compute an incremental cost-effectiveness ratio and assign it to the correct quadrant; recognise a dominated and a dominant alternative and say why a ratio alone cannot distinguish them; run a least-cost comparison over alternatives delivering the same outcome; state the outcome measure, its unit and its source; and write a cost-effectiveness case that says what it does not establish – including that it establishes nothing about whether the outcome is worth having.

What this course does not claim

This course does not tell you whether an outcome is worth its cost. That is precisely the question cost-effectiveness cannot answer: it compares the cost of achieving a given outcome across alternatives, and it is silent on whether the outcome should be pursued at all. That decision is political and the appraisal should say so rather than disguising it.

It does not supply outcome measures, burden estimates or effectiveness evidence. A cost per DALY averted requires a burden estimate somebody produced; a cost per additional completer requires an effectiveness study. This course states what each requires and where it comes from.

It does not replace economic appraisal where monetisation is defensible. Reaching for cost-effectiveness because a benefit-cost ratio came out badly is the same failure as monetising until the ratio clears – in the opposite direction.

It does not draw on stored cost-effectiveness results, because there are none. `project_costeffectiveness_results` holds 0 rows: the engine is implemented in 256 lines with 4 functions, and it has never been run against the register. Every ratio in this course is computed by hand from the implemented equations and the stored cost and output data, and the course says so wherever a number appears.

Learning outcomes

1. Decide whether an outcome can be defensibly monetised, and state what would have to be true for it to be.
2. Recognise the failure of monetising until a ratio clears, and its mirror image.
3. Compute a cost-effectiveness ratio on both discounted and undiscounted denominators.
4. Explain why both denominators are stored and how much the choice changes the ratio.
5. Compute an incremental cost-effectiveness ratio and assign it to a quadrant.
6. Recognise dominant and dominated alternatives and say why a ratio alone cannot distinguish them.
7. Run a least-cost comparison over alternatives delivering the same outcome.
8. Write a cost-effectiveness case that states it establishes nothing about whether the outcome is worth having.

Module structure

Module	Title	Sessions	Hours	Laboratory	Manual	Assessment
M01	What cannot be priced	1	1.25	–	1	Quiz M01
M02	The wrong responses, both of them	1	3.25	Lab 01	2	Quiz M02
M03	The cost-effectiveness ratio	1	1.50	–	3	Quiz M03
M04	Two denominators	1	3.50	Lab 02	4	Quiz M04
M05	The incremental ratio	1	1.50	–	5	Quiz M05
M06	The four quadrants	1	3.50	Lab 03	6	Quiz M06
M07	Least-cost comparison	1	3.50	Lab 04	7	Quiz M07
M08	Writing the case	1	1.50	–	8	Quiz M08

Laboratories

Lab	Title	Hours	Deliverable
1	Decide the instrument, and record the reason	2.0	Three instrument decisions with recorded reasons
2	Compute a cost-effectiveness ratio both ways	2.0	A ratio computed both ways, with the difference measured
3	Incremental ratios, and the quadrant that decides the reading	2.0	Four incremental ratios, four quadrants, four readings
4	Least-cost comparison, and the case that says what it does not establish	2.0	A least-cost comparison and a complete case

Assessment

Component	Weight
Knowledge quizzes	15 %
Laboratory exercises	30 %
Applied assignments	25 %

Component	Weight
Capstone / final project	30 %

Applied assignments

ID	Title	Weight	Deliverable
A1	The instrument decision	25	Marked on distinguishing evidence from policy obstacles
A2	The ratio, both ways	25	Marked on the factor and on which denominator drives the conclusion
A3	Quadrants and readings	25	Marked on the two that need no ratio and the ambiguous pair
A4	The cost-effectiveness case	25	Marked on the closing section and its named owner

Final project

A complete cost-effectiveness case for the register's non-monetised project: a justified statement that the outcome cannot be defensibly monetised and what would have to be true for it to be; the outcome measure with its unit and source; cost-effectiveness ratios on both denominators with the difference between them stated; at least two alternatives compared on an incremental ratio with its quadrant identified; a least-cost comparison; and a closing section stating that the analysis establishes nothing about whether the outcome is worth having, and who decides that.

Folder contents

Every course in the Academy carries the same section 5 folder. See `README.md` for what each item holds and `COURSE_BUILD_REPORT.md` for what was measured.